

DATA MINDS

THE FABRIC QUICKSTART GUIDE



From Zero to Insights:
Your 4-Week Roadmap to a Modern Data Estate

The Foundation: Setup & Licensing

Before we build, we need to prepare the ground. Microsoft Fabric simplifies infrastructure by relying on **Unified Capacity** rather than individual servers.

STEP 1: IGNITE YOUR ENGINE

You have two options to start without committing to a large budget:

- **Option A (Trial):** Activate the *Microsoft Fabric Trial* from your Power BI profile (60 days free, equivalent to F64 power).
- **Option B (Azure Pay-As-You-Go):** Purchase an F-SKU capacity. An **F2** or **F4** node is ideal for development and can be paused.

STEP 2: ONELAKE (YOUR ONEDRIVE FOR DATA)

Forget about managing scattered storage accounts. When you activate Fabric, you automatically get OneLake. It is a single, logical data lake for your entire organization.



OneLake

- A key component of Microsoft Fabric designed to be a unified, enterprise-wide data lake that simplifies the management of data across an organization



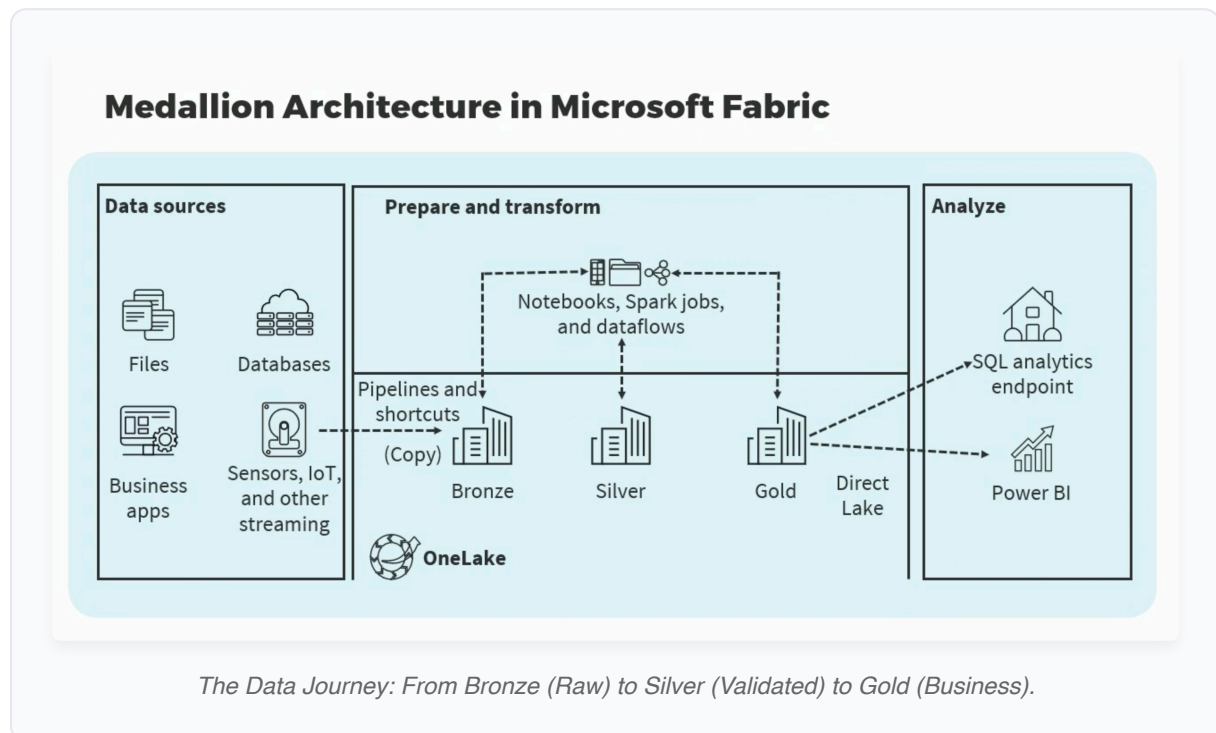
OneLake eliminates silos: A single copy of data, accessible by all engines.

STEP 3: WORKSPACE CONFIGURATION

Go to **app.fabric.microsoft.com** and create a new Workspace. ⚠️ **Crucial:** In the advanced settings, you must assign this workspace to your "Trial" or "Fabric Capacity" to unlock all features.

The Strategy: Medallion Architecture

Data quality is not an accident; it is an architecture. We do not dump messy data directly into an executive report. We use the industry standard:

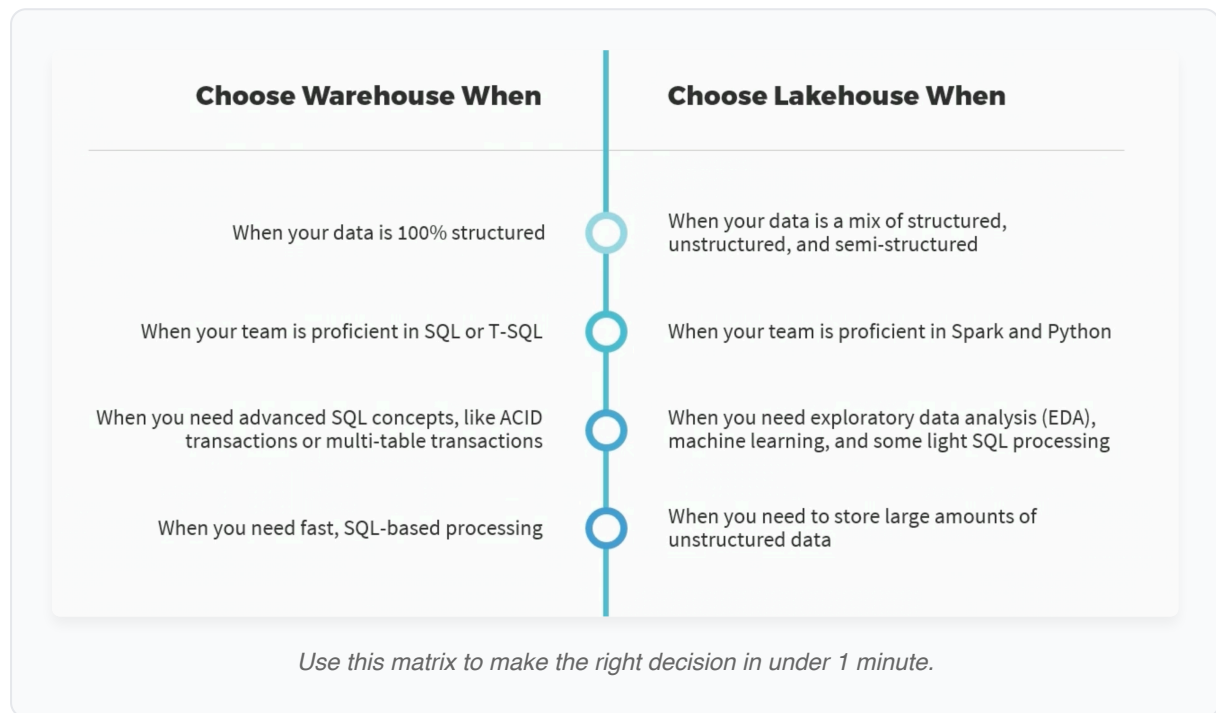


THE THREE LAYERS OF TRUTH

- **1 Bronze (Raw Layer):** Data landing zone. Exact, immutable copy of the source (JSON, CSV, Parquet).
- **2 Silver (Validated Layer):** Cleaning and standardization. Duplicates are removed, data types corrected, and nulls managed.
- **3 Gold (Enriched Layer):** Modeled for consumption. Here we build the Star Schema with business metrics ready for Power BI.

The Engine: Warehouse vs. Lakehouse

Fabric offers two main paths for storing and processing data. Choosing the right one at the start will save weeks of refactoring.



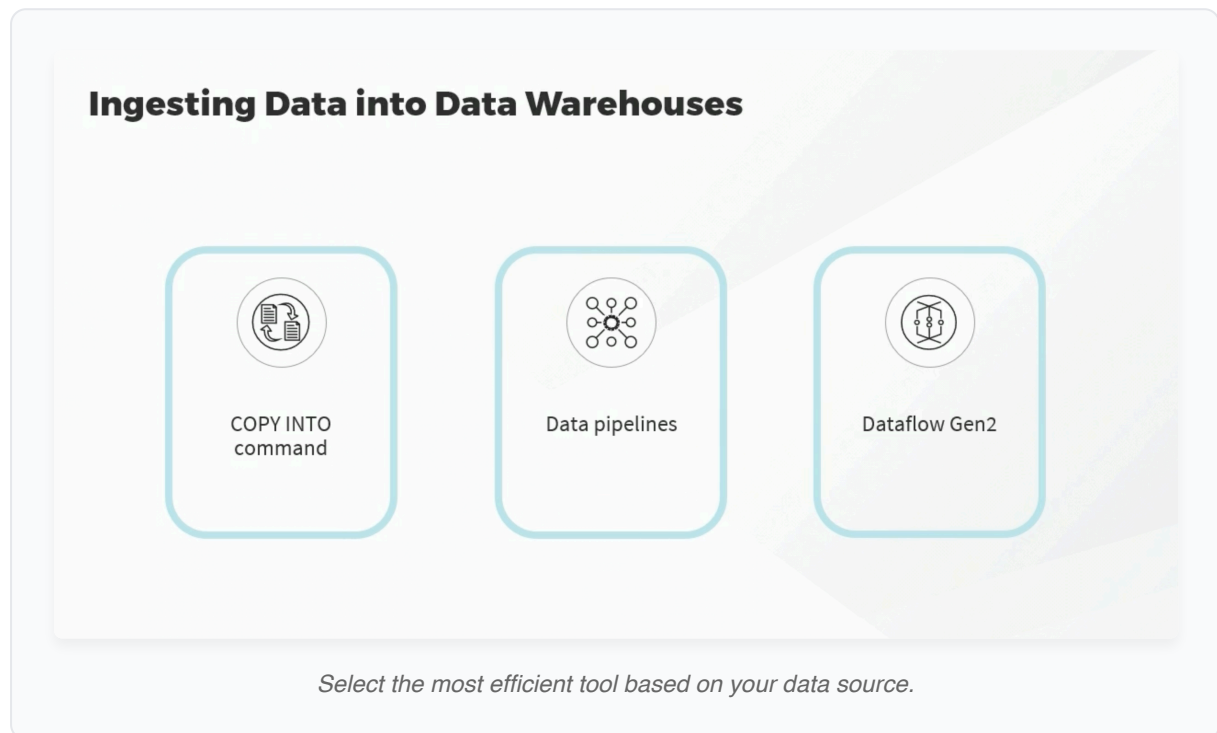
THE GOLDEN RULE

Choose Data Warehouse if: Your team consists of traditional SQL developers, your data is highly structured (ERP, CRM), and you need complex multi-table transactions.

Choose Data Lakehouse if: You want total flexibility, have semi-structured files (logs, images, complex JSON), or plan to use Python/Spark for Data Science.

Step 1: Ingestion (To Bronze)

The first technical step is landing data into the OneLake Bronze layer. You have three main tools at your disposal.

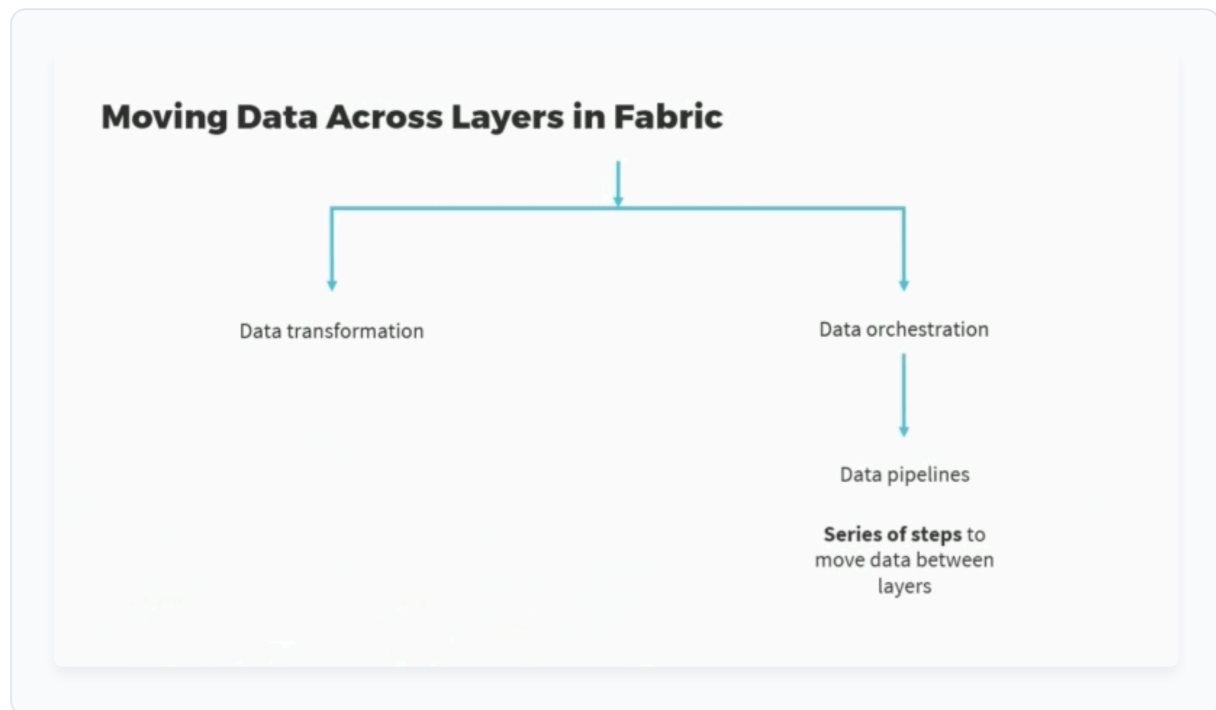


INGESTION METHODS

- **Data Pipelines (Copy Activity):** Ideal for massive, scheduled loads from On-Premise databases (via Gateway) or external clouds.
- **Shortcuts (The Fabric Advantage):** If your data is already in Azure (ADLS Gen2) or Amazon S3, **do not copy it**. Create a Shortcut to virtualize it instantly.
- **Dataflow Gen2:** Perfect for connecting to complex APIs, SharePoint lists, or local Excel files with visual transformation.

Step 2: Transformation (To Silver)

With data in Bronze, we apply cleaning logic. In Fabric, we clearly separate transformation (Logic) from orchestration (Timing).



OPTION A: DATAFLOWS GEN2 (LOW-CODE)

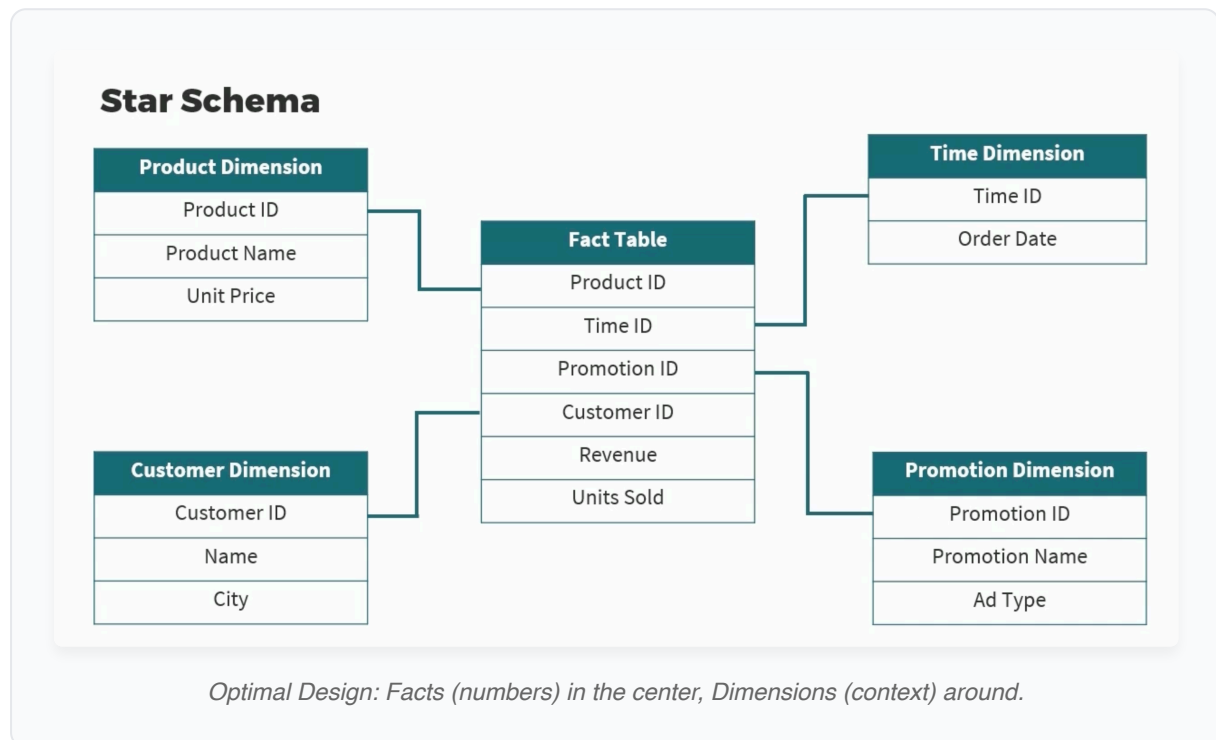
Ideal if you come from the Power BI world. Use the **Power Query Online** visual interface to remove nulls, filter rows, correct data types, and standardize column names without writing code.

OPTION B: NOTEBOOKS (PRO-CODE)

Ideal for Data Engineers. Use **PySpark** to read Bronze files, apply complex business logic or Python libraries, and write the result as optimized Delta tables.

Step 3: Modeling (The Gold Layer)

The #1 mistake in BI is connecting the report to a giant flat table. For real performance and flexibility, you need to build a **Star Schema**.

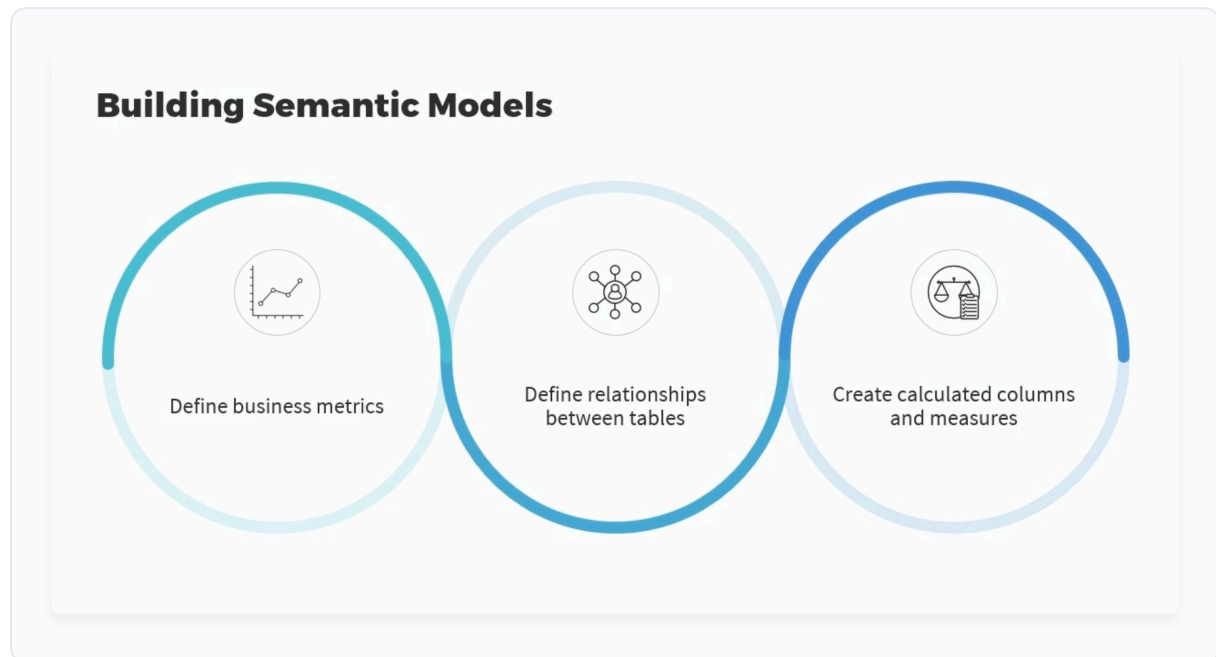


HOW TO BUILD IT:

1. **Identify Dimensions:** Descriptive tables (Who, When, Where). E.g., *Dim_Customer*, *Dim_Product*.
2. **Identify Facts:** Transactional tables (What happened). E.g., *Fact_Sales*.
3. **Relate:** Ensure "One-to-Many" relationships flow from Dimension to Fact.

Step 4: Semantic Model & Logic

Centralize your business logic in the semantic model, never in scattered reports.



DIRECT LAKE: EXTREME SPEED

Thanks to Fabric integration, create your semantic model by selecting Gold layer tables. Power BI will read data directly from OneLake (Direct Lake mode), eliminating import latency.

Step 5: Automation

"Set and forget" by creating a **Master Pipeline** that orchestrates the entire flow daily:



Execution Sequence:

1. Ingestion (Source → Bronze)
 2. Cleaning (Bronze → Silver)
 3. Modeling (Silver → Gold)
 4. Semantic Model Refresh
- Schedule: Daily at 06:00 AM*

The QuickStart Checklist

Before launching to production, verify that your environment meets quality standards:

- ☐ Workspace assigned to an active Fabric capacity.
- ☐ Data ingested into Bronze (raw state).
- ☐ Data cleaned in Silver (Delta format).
- ☐ Star Schema validated in Gold.
- ☐ Semantic Model with explicit DAX measures.
- ☐ Power BI report connected via Direct Lake.
- ☐ Orchestration pipeline scheduled and tested.

STUCK ON STEP 3?

Architecture is easy to draw but hard to implement perfectly.
We help Australian businesses build this foundation in weeks.

Book a 30-Min Strategy Call

Visit dataminds.au or email hello@dataminds.au